Assessment Menu: Teacher Guide

The Polynomial Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
OPERATE	Add, subtract and multiply polynomials	A1, A2, E1, E5, S1
GCF	Greatest common factor	A3, E2, S4
FACTOR	Factor trinomials and special products	A4, A5, E3, S2
SOLVE	Solve equations by factoring	A6, E4, E6, S3

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 OPERATE	Add: $(3x^2 + 5x - 2) + (x^2 - 7x + 6)$	**Answer:** $4x^2 - 2x + 4$ Combine like terms: $3x^2 + x^2 = 4x^2$; $5x - 7x = -2x$; $-2 + 6 = 4$	2 pts: Combines the x^2 terms 2 pts: Combines the x terms 1 pts: Constant 4
A2 OPERATE	Multiply: $(x + 4)(x - 3)$	**Answer:** $x^2 + x - 12$ $x^2 - 3x + 4x - 12$ Combine: $x^2 + x - 12$	2 pts: Four products correct (FOIL or box) 2 pts: Combines $-3x + 4x = x$ 1 pts: Final answer $x^2 + x - 12$
A3 GCF	Factor out the greatest common factor: $12x^3 + 18x^2$	**Answer:** $6x^2(2x + 3)$ GCF of 12 and 18 is 6; the smallest power of x is x^2, so the GCF is $6x^2$. $12x^3 + 18x^2 = 6x^2(2x + 3)$	2 pts: GCF $6x^2$ 2 pts: Correct quotients $2x$ and 3 1 pts: Final factored form
A4 FACTOR	Factor: $x^2 + 7x + 12$	**Answer:** $(x + 3)(x + 4)$ Find two numbers with product 12 and sum 7: 3 and 4. $(x + 3)(x + 4)$	3 pts: Identifies 3 and 4 2 pts: Correct factored form
A5 FACTOR	Factor: $4x^2 - 25$	**Answer:** $(2x - 5)(2x + 5)$ This is a difference of squares: $(2x)^2 - 5^2$. $(2x - 5)(2x + 5)$	2 pts: Recognizes a difference of squares 2 pts: Square roots $2x$ and 5 1 pts: Correct signs in the two factors
A6 SOLVE	Solve by factoring: $x^2 - 5x - 14 = 0$	**Answer:** $x = 7$ or $x = -2$ $(x - 7)(x + 2) = 0$ $x - 7 = 0$ or $x + 2 = 0$, so $x = 7$ or $x = -2$	2 pts: Factors correctly 1 pts: Sets each factor equal to 0 2 pts: Both solutions

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 OPERATE	Multiply and simplify: $(2x - 3)(x^2 + 4x - 5)$	**Answer:** $2x^3 + 5x^2 - 22x + 15$ $2x(x^2 + 4x - 5) = 2x^3 + 8x^2 - 10x$ $-3(x^2 + 4x - 5) = -3x^2 - 12x + 15$ Combine: $2x^3 + 5x^2 - 22x + 15$	3 pts: Distributes $2x$ correctly 3 pts: Distributes -3 correctly (watch signs) 2 pts: Combines like terms 2 pts: Final answer in standard form
E2 GCF	Factor completely: $3x^3 - 12x^2 - 36x$	**Answer:** $3x(x - 6)(x + 2)$ GCF is $3x$: $3x(x^2 - 4x - 12)$ $x^2 - 4x - 12 = (x - 6)(x + 2)$ $3x(x - 6)(x + 2)$	3 pts: Factors out $3x$ 4 pts: Factors the trinomial correctly 3 pts: Includes the GCF in the final answer
E3 FACTOR	Factor: $6x^2 + 11x - 10$. Show how you check your answer by multiplying.	**Answer:** $(3x - 2)(2x + 5)$ $ac = -60$; find two numbers with product -60 and sum 11: 15 and -4. $6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$ Check: $6x^2 + 15x - 4x - 10 = 6x^2 + 11x - 10$ ✓	3 pts: Finds 15 and -4 (or equivalent method) 3 pts: Splits and groups correctly 2 pts: Correct factors 2 pts: Checks by multiplying
E4 SOLVE	Solve: $2x^2 + 5x = 12$	**Answer:** $x = \frac{3}{2}$ or $x = -4$ Set equal to 0: $2x^2 + 5x - 12 = 0$ Factor: $(2x - 3)(x + 4) = 0$ $2x - 3 = 0$ gives $x = \frac{3}{2}$; $x + 4 = 0$ gives $x = -4$	3 pts: Rewrites with 0 on one side 3 pts: Factors correctly 4 pts: Both solutions
E5 OPERATE	Lena subtracted: $(5x^2 + 2x - 1) - (3x^2 - 4x + 6) = 2x^2 - 2x + 5$ Find her mistake, explain it, and give the correct answer.	**Answer:** She did not subtract every term of the second polynomial. Correct: $2x^2 + 6x - 7$. The minus sign must distribute to all three terms: $-3x^2 + 4x - 6$. $5x^2 - 3x^2 = 2x^2$; $2x + 4x = 6x$; $-1 - 6 = -7$ Correct answer: **$2x^2 + 6x - 7$**	4 pts: Identifies the error: the negative was not distributed 3 pts: Explains with signs ($-(-4x) = +4x$, -6) 3 pts: Correct answer $2x^2 + 6x - 7$
E6 SOLVE	Solve each by factoring: **(a)** $3x^2 - 27 = 0$ **(b)** $x^2 + 6x + 9 = 0$ (Why does (b) give only one solution?)	**Answer:** (a) $x = 3$ or $x = -3$ (b) $x = -3$ (repeated) (a) $3(x^2 - 9) = 3(x - 3)(x + 3) = 0$, so $x = 3$ or $x = -3$ (b) $(x + 3)(x + 3) = 0$. Both factors are identical, so there is just one solution, $x = -3$.	2 pts: (a) Factors out 3 2 pts: (a) Both solutions 2 pts: (b) Factors as $(x + 3)^2$ 2 pts: (b) $x = -3$ 2 pts: Explains the repeated factor gives one solution

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 OPERATE	A rectangular pool is $(2x + 3)$ m long and $(x + 4)$ m wide. A deck 1 m wide surrounds the pool on all four sides. **(a)** Write the pool’s area as a polynomial. **(b)** Write the area of pool plus deck. **(c)** Find the deck’s area. **(d)** Evaluate (c) for $x = 4$.	**Answer:** (a) $2x^2 + 11x + 12$ (b) $2x^2 + 17x + 30$ (c) $6x + 18$ (d) 42 m² (a) $(2x + 3)(x + 4) = 2x^2 + 11x + 12$ (b) The deck adds 2 m to each dimension: $(2x + 5)(x + 6) = 2x^2 + 17x + 30$ (c) Subtract: $(2x^2 + 17x + 30) - (2x^2 + 11x + 12) = 6x + 18$ (d) $6(4) + 18 = 42$ m²	4 pts: (a) Correct pool area 3 pts: (b) Dimensions $2x + 5$ and $x + 6$ 3 pts: (b) Correct product 4 pts: (c) Subtracts with correct signs: $6x + 18$ 3 pts: (d) Correct evaluation 42 3 pts: Units (m²) and clear labeling
S2 FACTOR	Mia factored $4x^2 - 36$ as $(2x - 6)(2x + 6)$ and says she is done. **(a)** Show her answer multiplies back correctly. **(b)** Explain why it is not factored completely, and factor completely. **(c)** Factor completely: $2x^3 - 50x$.	**Answer:** (b) $4(x - 3)(x + 3)$ (c) $2x(x - 5)(x + 5)$ (a) $(2x - 6)(2x + 6) = 4x^2 + 12x - 12x - 36 = 4x^2 - 36$ ✓ (b) Each factor still has a common factor of 2, so $(2x - 6)(2x + 6) = 2(x - 3) \cdot 2(x + 3) =$ **$4(x - 3)(x + 3)$**. Better: factor out the GCF 4 first. (c) GCF $2x$: $2x(x^2 - 25) =$ **$2x(x - 5)(x + 5)$**	4 pts: (a) Multiplies back correctly 4 pts: (b) Explains that each factor still has a common factor of 2 4 pts: (b) Correct complete factorization $4(x - 3)(x + 3)$ 4 pts: (c) Factors out $2x$ 4 pts: (c) Factors the difference of squares
S3 SOLVE	A ball is thrown upward from an 80-foot cliff. Its height is $h = -16t^2 + 64t + 80$ feet after t seconds. **(a)** Factor the right side completely. **(b)** When does the ball hit the ground ($h = 0$)? Explain why one answer is rejected. **(c)** Besides $t = 0$, when is the ball again at 80 feet?	**Answer:** (a) $-16(t - 5)(t + 1)$ (b) 5 seconds (c) $t = 4$ seconds (a) $-16(t^2 - 4t - 5) =$ **$-16(t - 5)(t + 1)$** (b) $h = 0$ gives $t = 5$ or $t = -1$. Time cannot be negative, so the ball lands at **5 seconds**. (c) $80 = -16t^2 + 64t + 80$ gives $-16t^2 + 64t = 0$, so $-16t(t - 4) = 0$ and **$t = 4$ seconds**.	3 pts: (a) Factors out -16 3 pts: (a) Factors the trinomial 3 pts: (b) Sets each factor to 0: $t = 5$, $t = -1$ 3 pts: (b) Rejects -1 because time is not negative 4 pts: (c) Sets $h = 80$ and simplifies to $-16t^2 + 64t = 0$ 4 pts: (c) Solves: $t = 4$ seconds
S4 GCF	A rectangle has area $6x^3 + 15x^2 - 9x$ square units. Its width is the GCF of the three terms. **(a)** Find the width and the length. **(b)** Factor the length to write the area as a product of three factors. **(c)** Check with $x = 2$.	**Answer:** (a) width $3x$, length $2x^2 + 5x - 3$ (b) $3x(2x - 1)(x + 3)$ (c) area 90 (a) GCF is $3x$: $6x^3 + 15x^2 - 9x = 3x(2x^2 + 5x - 3)$. Width $3x$, length $2x^2 + 5x - 3$. (b) $2x^2 + 5x - 3 = (2x - 1)(x + 3)$, so area = **$3x(2x - 1)(x + 3)$** (c) $x = 2$: $6(8) + 15(4) - 18 = 90$ and $3(2)(3)(5) = 90$ ✓	4 pts: GCF $3x$ 4 pts: Length $2x^2 + 5x - 3$ 5 pts: Factors the length as $(2x - 1)(x + 3)$ 3 pts: Writes the three-factor product 4 pts: (c) Evaluates both forms to 90